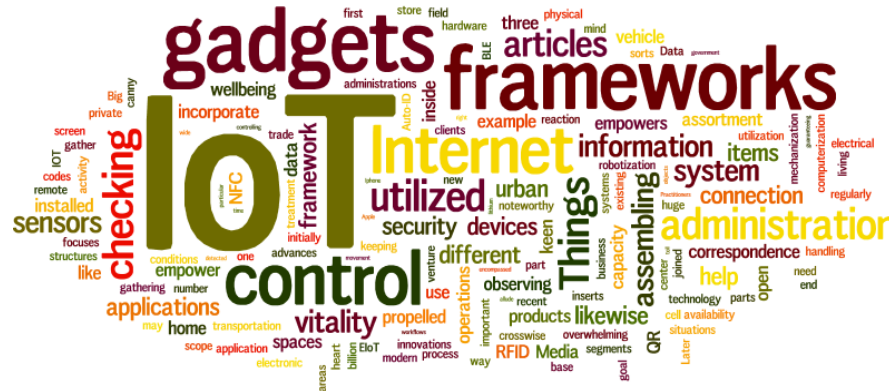


CS578: Internet of Things

IEEE 802.15.4

Standard: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=6471722>



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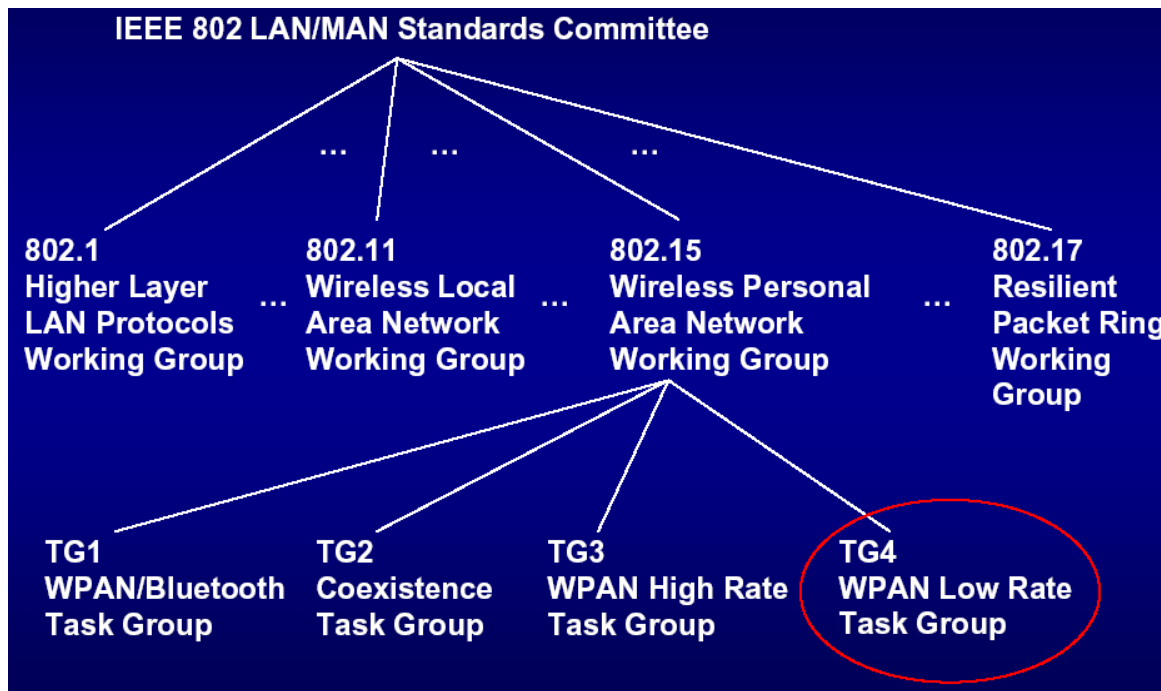
“The highest education makes our life in harmony with all existence.” – Rabindranath Tagore

IEEE 802.15 Task Group 4

- TG4 defines **low-data-rate PHY and MAC layer specifications** for wireless **personal area networks** (WPAN)
- This standard has evolved over the years:
 - IEEE 802.15.4-2003 ; IEEE 802.15.4-2006
 - IEEE 802.15.4-2011; **IEEE 802.15.4-2015**

PAN

- span a **small area** (e.g., a private home or an individual workspace)
- communicate over a **short distance**
- **low-powered** communication
- primarily uses **ad-hoc networking**
- could be **wireless** or **wired** (e.g. using USB)



IEEE 802.15.4 market feature



- Low **power** consumption
 - Low **cost** system and operation
 - Low offered message **throughput**
 - Supports large **network** ($\leq 65k$ nodes)
 - Low to no **QoS** guarantees
 - Flexible protocol **design**
-
- IEEE 802.15.4 PHY and MAC layers **are the foundations** for several networking protocol stacks used in different market applications.
 - Few well-known protocol stacks:
 - ZigBee
 - ZigBee IP
 - 6LoWPAN
 - WirelessHART
 - Thread
 - **ZigBee** shows how 802.15.4 can be leveraged at the PHY and MAC layers, independent of the protocol layers above.

What is ZigBee Alliance?

- An **alliance** of organizations with a **mission** to define
 - reliable,
 - cost effective,
 - low-power,
 - wirelessly networked,
 - monitoring and control products
 - based on an open global standard
- Alliance provides
 - interoperability,
 - certification testing, and
 - branding

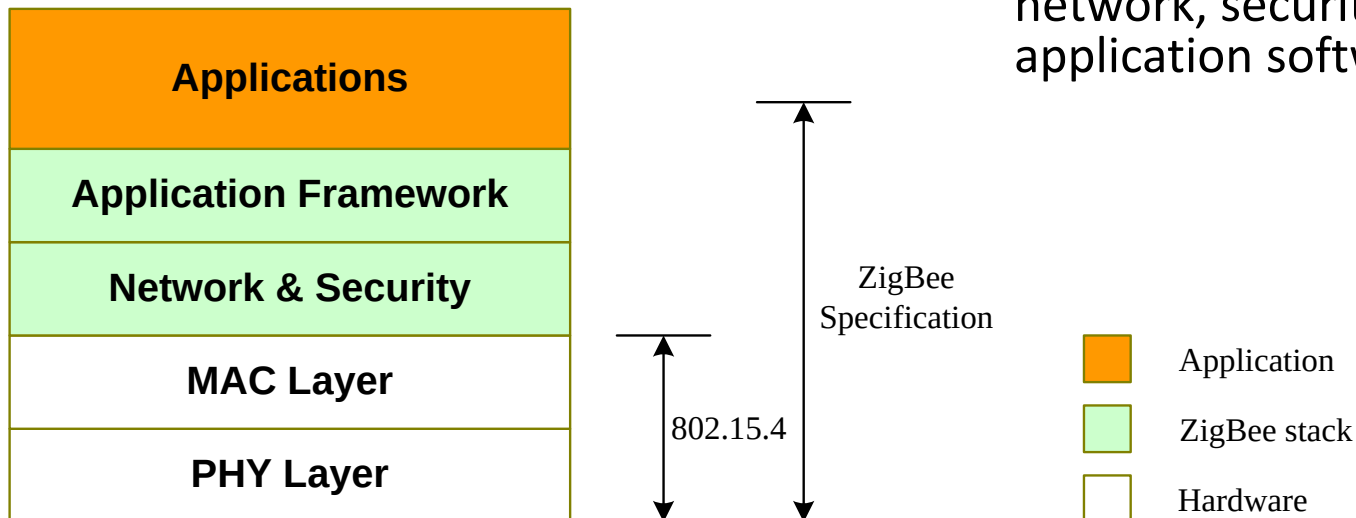


- ZigBee Alliance
 - 45+ companies: Semiconductor mfrs, IP providers, OEMs, etc.

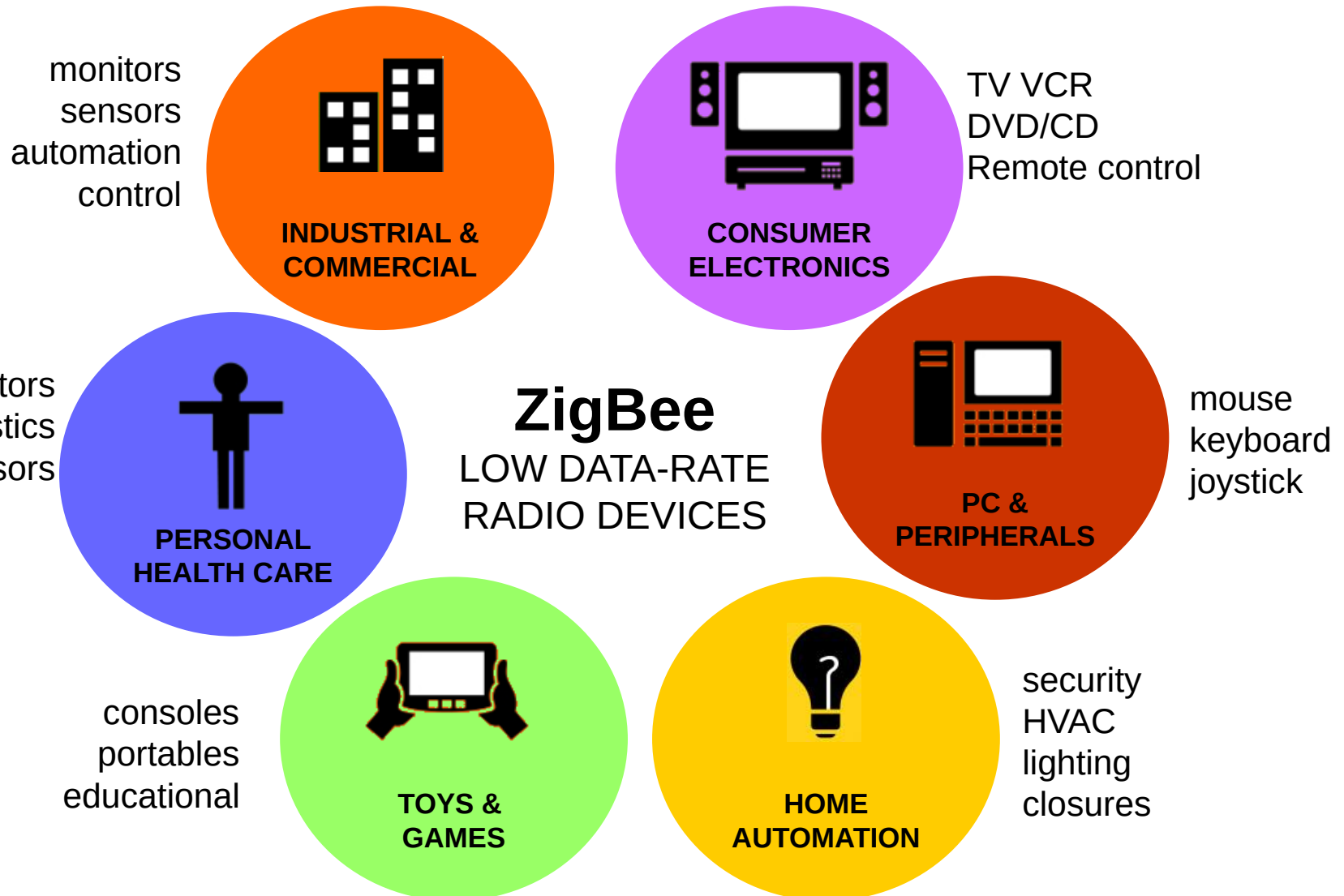
ZigBee/802.15.4 architecture

- ZigBee Alliance
 - Defining upper layers of protocol stack: from network to application, including application profiles
- IEEE 802.15.4 Working Group
 - Defining lower layers of protocol stack: MAC and PHY

- ZigBee takes full advantage of a powerful physical radio specified by IEEE 802.15.4
- ZigBee adds logical network, security and application software



ZigBee network applications



IEEE 802.15.4 PHY

IEEE 802.15.4 PHY overview



- PHY functionalities:
 - Activation & deactivation of the radio transceiver
 - Energy detection within the current channel
 - Link quality indication for received packets
 - Clear channel assessment (CCA) for CSMA-CA
 - Channel frequency selection
 - Data transmission and reception

Spectrum

- Federal Communications of Commissions (FCC) in USA allocates frequency bands
- Applications using ISM band do not require a licence for **stations emitting less than 1W**.

FCC Band	Max. Transmit Power	Frequencies
Industrial Band	< 1 W	902 MHz – 928 M Hz
Scientific Band	< 1 W	2.4 GHz – 2.48 GHz
Medical Band	< 1 W	5.725 GHz – 5.85 GHz
U-NII (Unlicensed National Information Infrastructure)	< 40 mW	5.15 GHz – 5.25 GHz
	< 200 mW	5.25 GHz – 5.35 GHz
	< 800 mW	5.725 GHz – 5.82 GHz

- Physical layer **transmission options** in IEEE 802.15.4-2015
 - **2.4 GHz**, 16 channels, data rate 250 kbps
 - **915 MHz**, 10 channels, data rate 250 kbps
 - **868 MHz**, 3 channel, data rate 100 kbps

Spread Spectrum

- Idea of **Spread Spectrum** is to **spread the information signal** over a wider bandwidth to **make jamming and interception more difficult**.
 - can be used to transmit **analog / digital data**, **using an analog signal**
- Types:**
 - frequency hopping spread spectrum (**FHSS**)
 - direct sequence spread spectrum (**DSSS**)

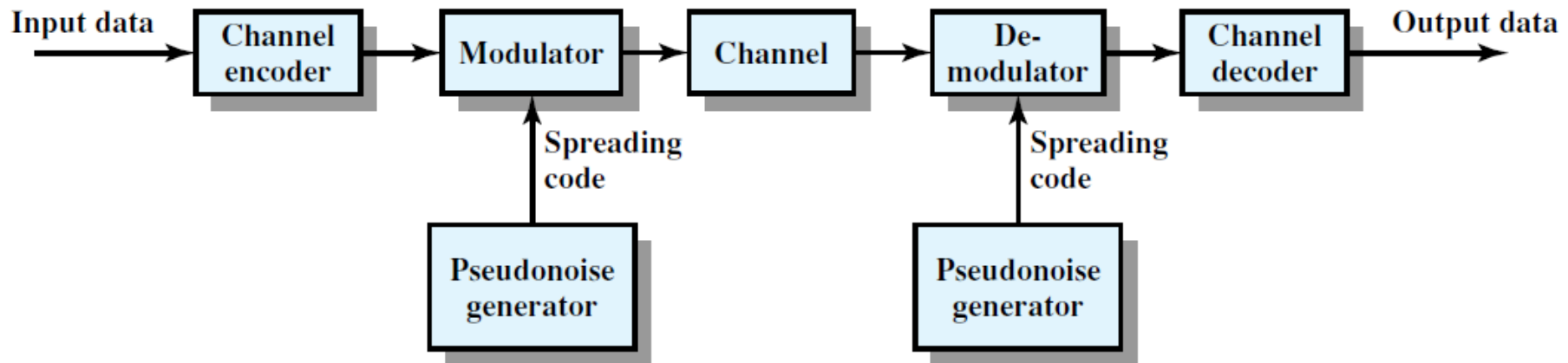


Figure 9.1 General Model of Spread Spectrum Digital Communication System

Channel coding is accomplished by selectively introducing redundant bits into the transmitted information stream. **Modulation** is the process by which some characteristic of a carrier wave is varied in accordance with an information signal, or modulating wave.

Cont...



- Pseudorandom numbers
 - generated by an algorithm using some initial value called the **seed**
 - produce sequences of numbers that are **not statistically random**, but passes reasonable tests of randomness
 - unless you know the **algorithm** and the **seed**, **it is impractical to predict the sequence**
- **Gain from** this **apparent waste of spectrum**
 - The signals **gains immunity** from various kinds of noise and multipath distortion.
 - **Immune to** jamming attack
 - It can also be used for **hiding and encrypting signals**.
 - **Several users can independently use** the same higher bandwidth with very little interference. (e.g. CDMA)

- each bit in the original signal is represented by multiple bits in the transmitted signal, using a spreading code
- spreading code spreads the signal across a wider frequency band in direct proportion to the number of bits used

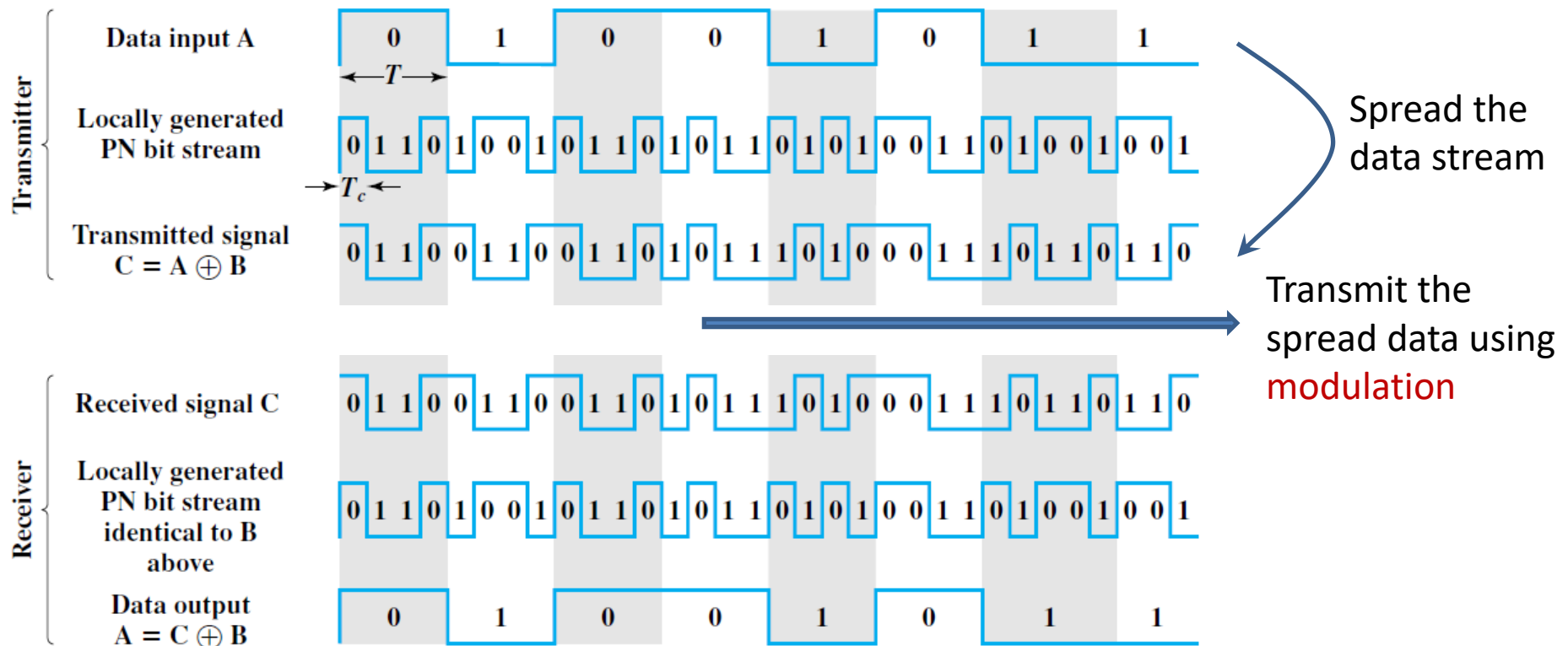


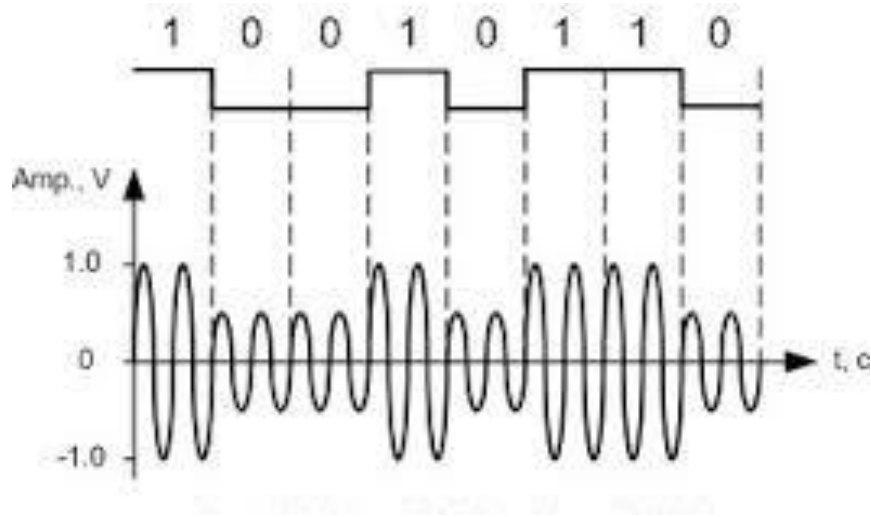
Figure 9.6 Example of Direct Sequence Spread Spectrum

Modulation

Modulation schemes

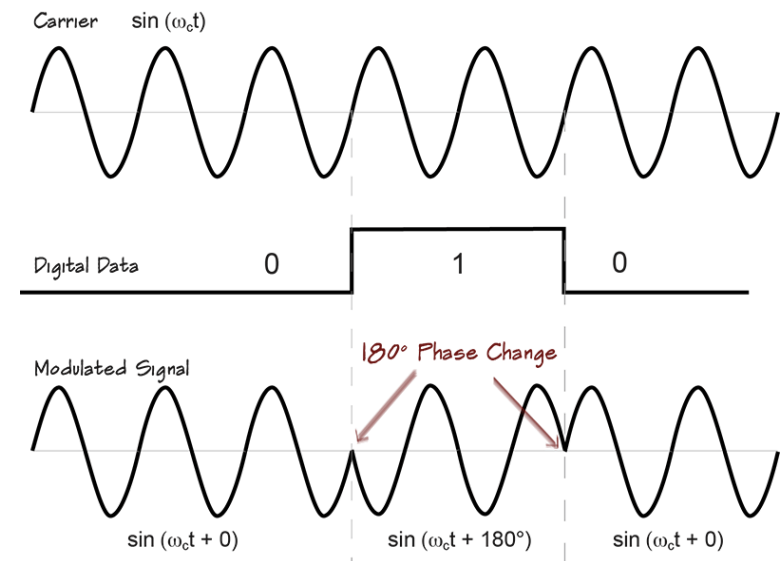
- **OQPSK PHY** : DSSS PHY employing Offset Quadrature Phase-Shift Keying (OQPSK)
- **BPSK PHY** : DSSS PHY employing binary phase-shift keying (BPSK)
- **ASK PHY** : PSSS PHY employing Amplitude Shift Keying (ASK) and BPSK

$$\text{ASK } s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ 0 & \text{binary 0} \end{cases}$$



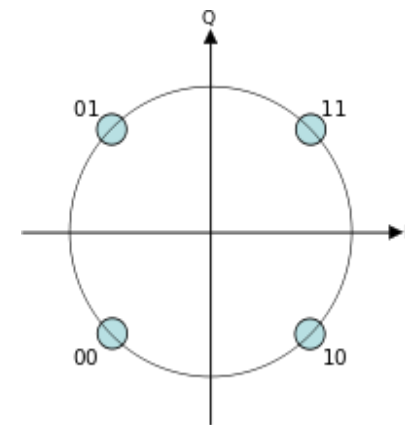
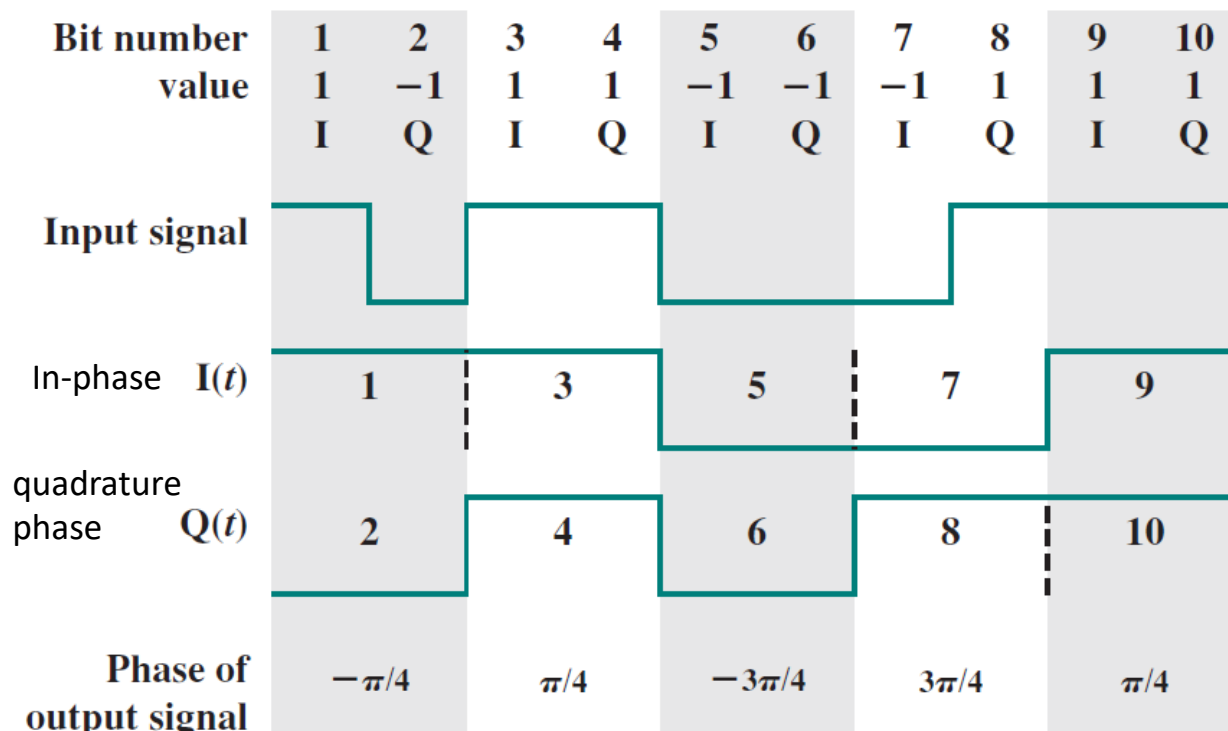
Amplitude Shift Keying (ASK)

$$\text{BPSK } s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ A \cos(2\pi f_c t + \pi) & \text{binary 0} \end{cases} = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ -A \cos(2\pi f_c t) & \text{binary 0} \end{cases}$$



Binary Phase-Shift Keying (BPSK)

QPSK



Constellation diagram for QPSK

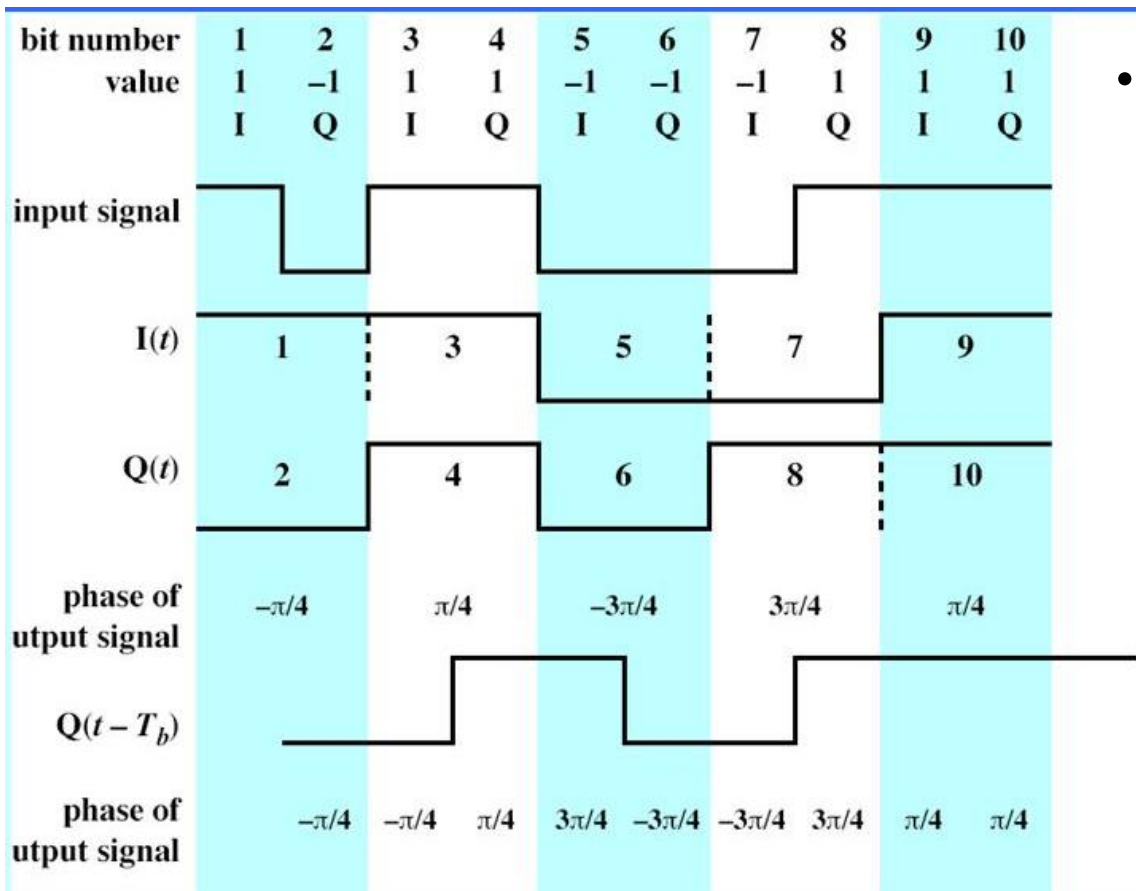
Quadrature Phase-Shift Keying (QPSK)

- More efficient use of bandwidth
 - as each signalling element represents more than one bit.

$$\text{QPSK } s(t) = \begin{cases} A \cos\left(2\pi f_c t + \frac{\pi}{4}\right) & 11 \\ A \cos\left(2\pi f_c t + \frac{3\pi}{4}\right) & 01 \\ A \cos\left(2\pi f_c t - \frac{3\pi}{4}\right) & 00 \\ A \cos\left(2\pi f_c t - \frac{\pi}{4}\right) & 10 \end{cases}$$

Orthogonal QPSK

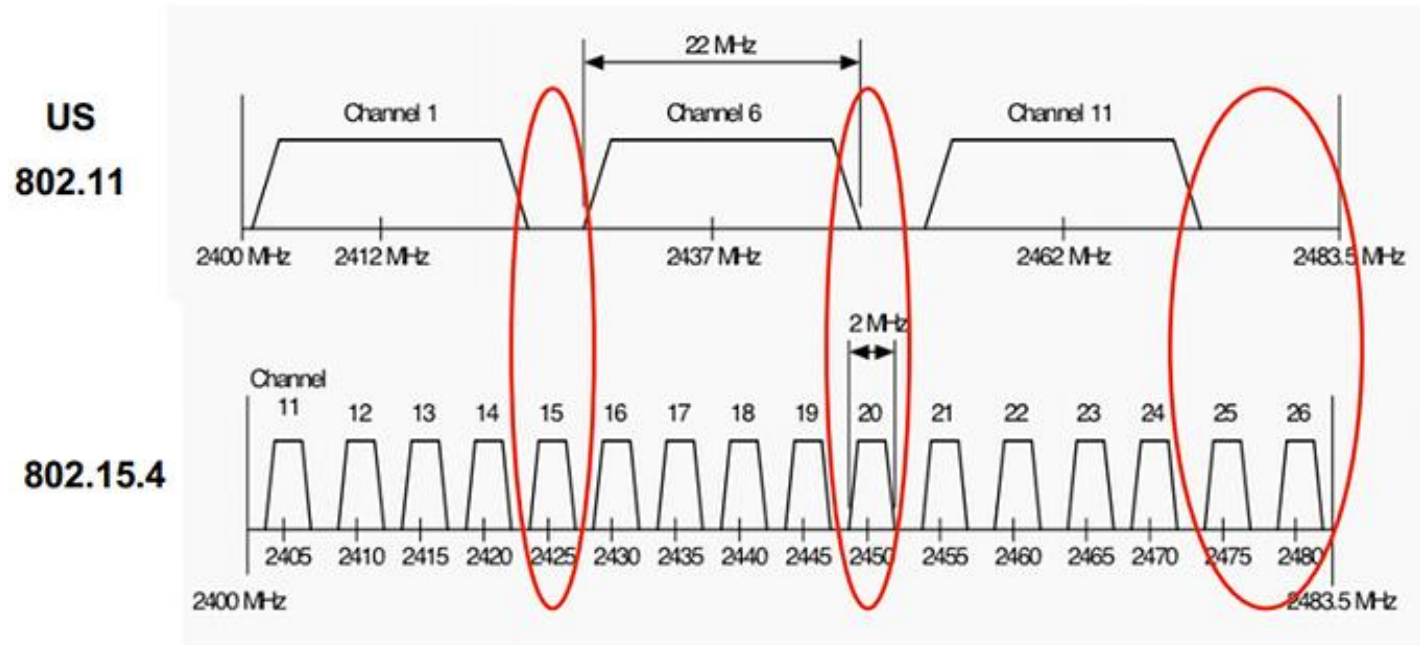
- **Problem in QPSK:** large phase shift at high transition rate is difficult to perform. Phase shift is 180° in QPSK.



• OQPSK

- ✓ a variation of QPSK known as **offset QPSK** or **orthogonal QPSK**
- ✓ a **delay of one bit time** is introduced in the Q stream of QPSK
- ✓ Its spectral characteristics and bit-error performance are the same as that of QPSK
- ✓ at any time the **phase change** in the combined signal **never exceeds 90° ($\pi/2$)**

Other Attributes



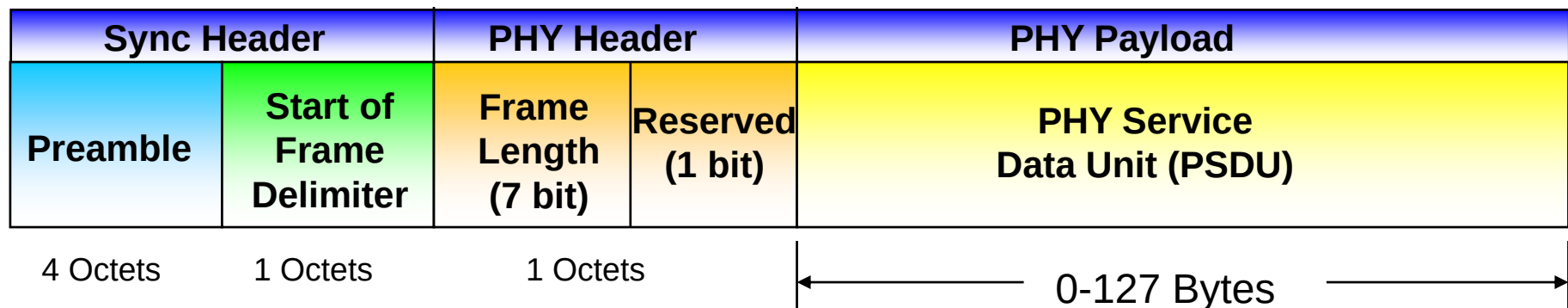
- IEEE 802.15.4 **does not prefer to use frequency hopping** to **minimize energy consumption**.
- To minimize interference in 2.4 GHz band, IEEE 802.15.4 prefer **channel no. 15, 20, 25, 26**
- Transmission power is adjustable from 0.5 mW (min in 802.15.4) to 1 W (max in ISM band)
- Transmission power 1 mW provides **theoretical distances** as:
 - Outdoor range **300 m**.
 - Indoor range **100 m**.

- 802.15.4 PHY provides **energy detection (ED)** feature
 - **Application** can request to **asses** each channel's **energy level**
 - **Coordinator** can make **optimal selection of channel** based on channels energy level
- 802.15.4 PHY provides **link quality information (LQI)** to **NET and APP layers**
 - Transmitter may **decide to use high transmission power** based on LQI
 - Applications may **dynamically change 802.15.4 channels** based on LQI
- 802.15.4 uses **CSMA/CA** which ask the PHY layer to do CCA
 - **Clear Channel Assessment (CCA)**:
 - Can be energy threshold regardless of modulation
 - Can be detection of modulation
 - Can be both the above

PHY Frame



- PHY packet fields
 - Preamble** (32 bits) – synchronization of data transmission
 - SFD** (8 bits) – shall be formatted as “1110 0101”
 - PHY header** (8 bits) – PSDU length
 - PSDU** (0 to 127 bytes) – data field

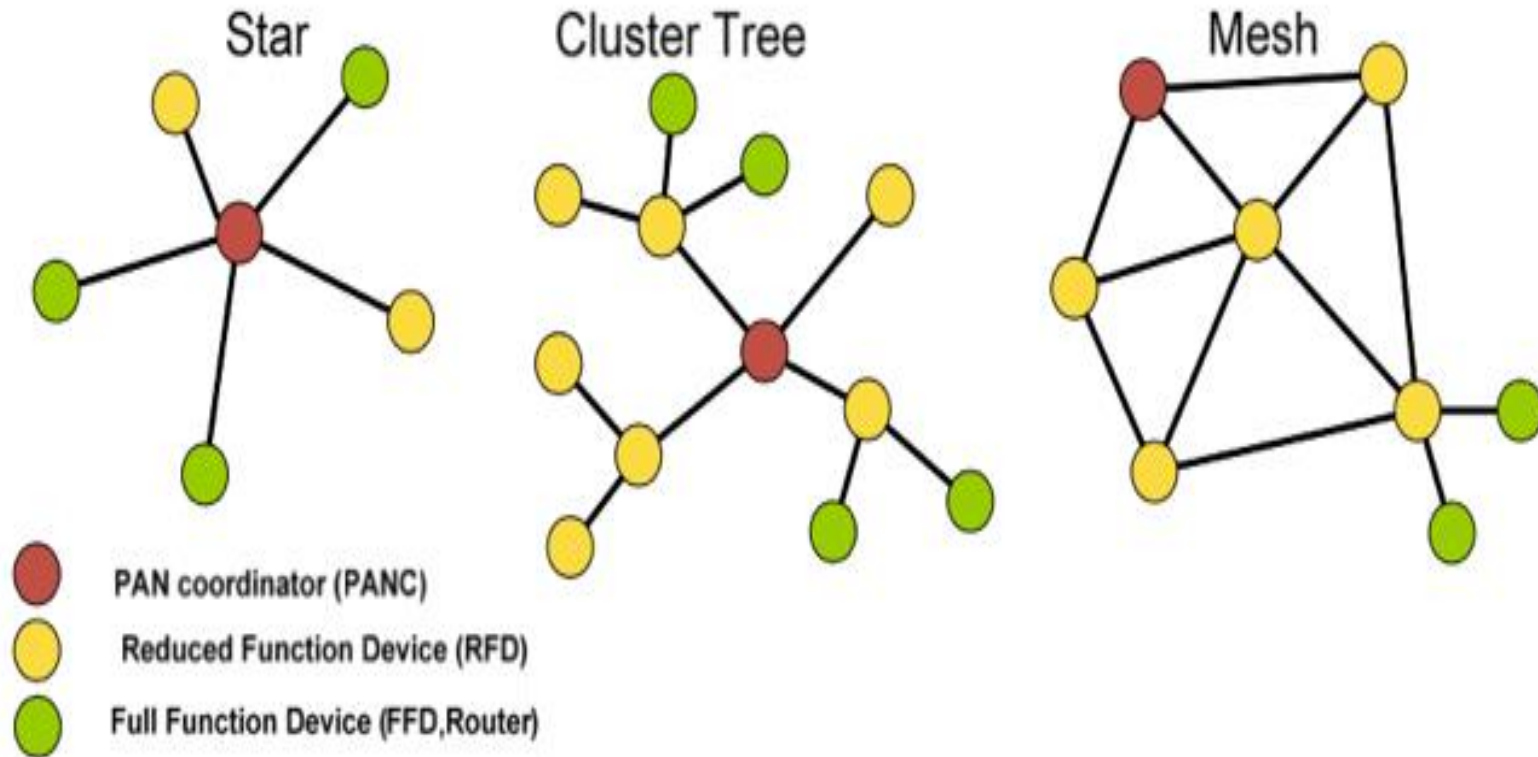


IEEE 802.15.4 MAC

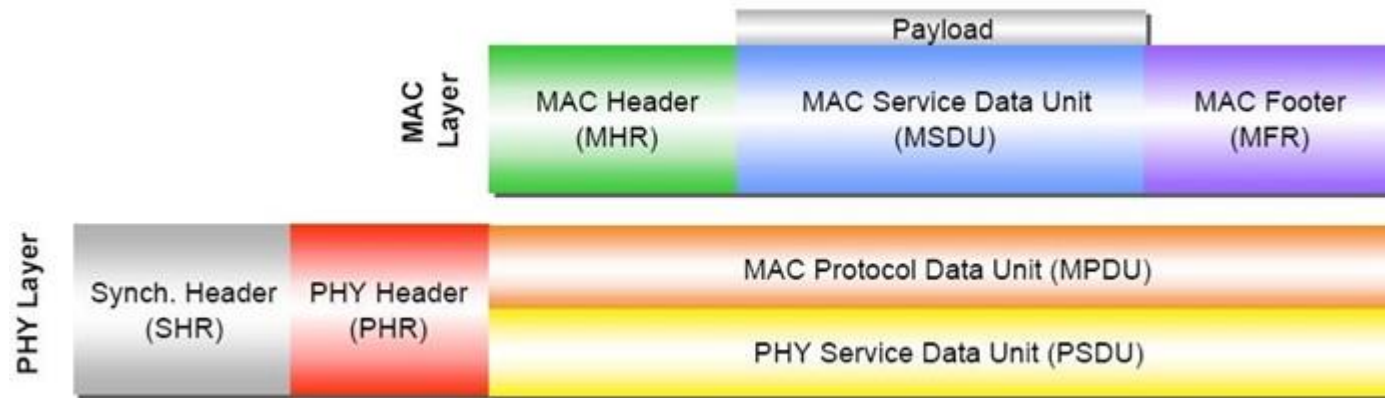
IEEE 802.15.4 Device Types

- There are **two different device types** :
 - full function device (**FFD**)
 - reduced function device (**RFD**)
- The **FFD** can operate in **three modes** by serving as
 - **PAN Coordinator**
 - scanning the network and selecting optimal RF channel
 - selecting the 16 bit PAN ID for the network
 - **Coordinator (aka Parent, Join Proxy)**
 - relaying messages to other FFDs including PAN coordinator
 - transmits periodic beacon (under beacon enable access mode)
 - respond to beacon requests
 - **Device**
 - cannot route messages
 - usually receivers are switched off except during transmission
 - attached to the network only as leaf nodes
- The **RFD** can only serve as:
 - **Device**

Topology



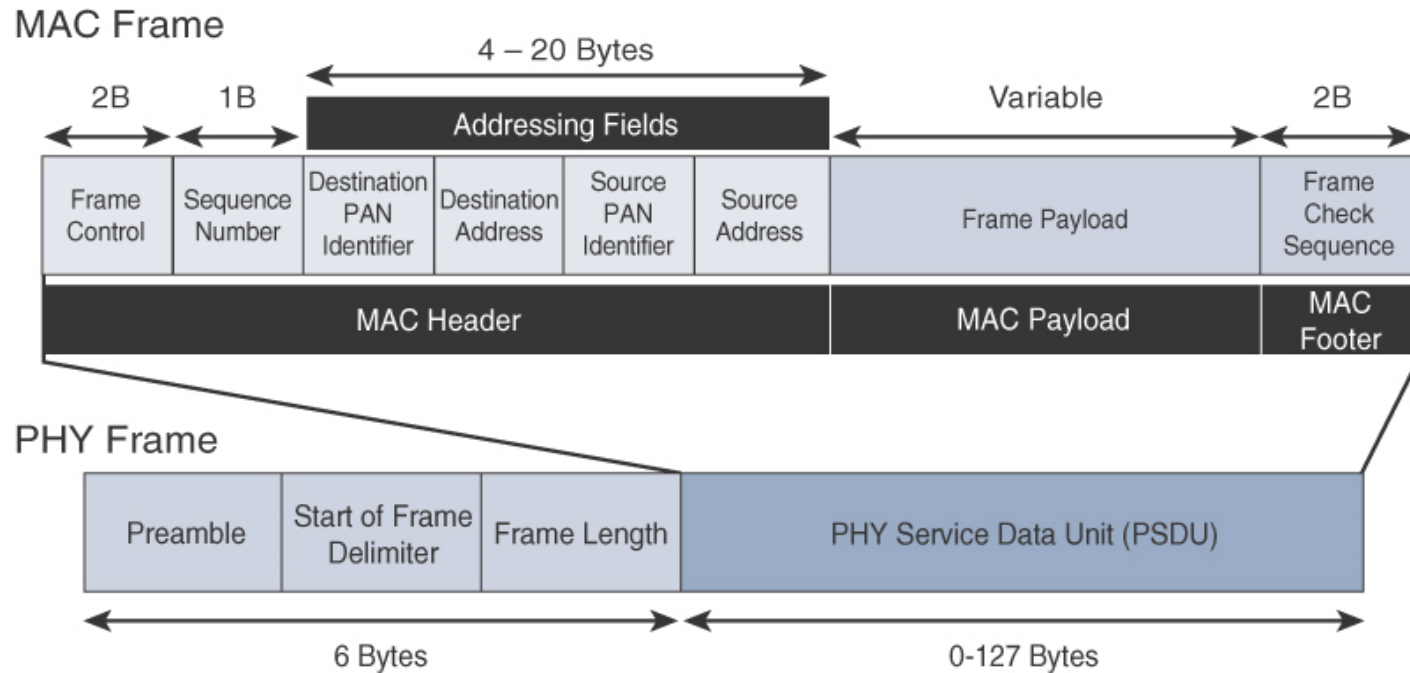
General MAC Frame Format



MAC frame types:

- Data frame
- ACK frame
- Beacon frame
- Command frame

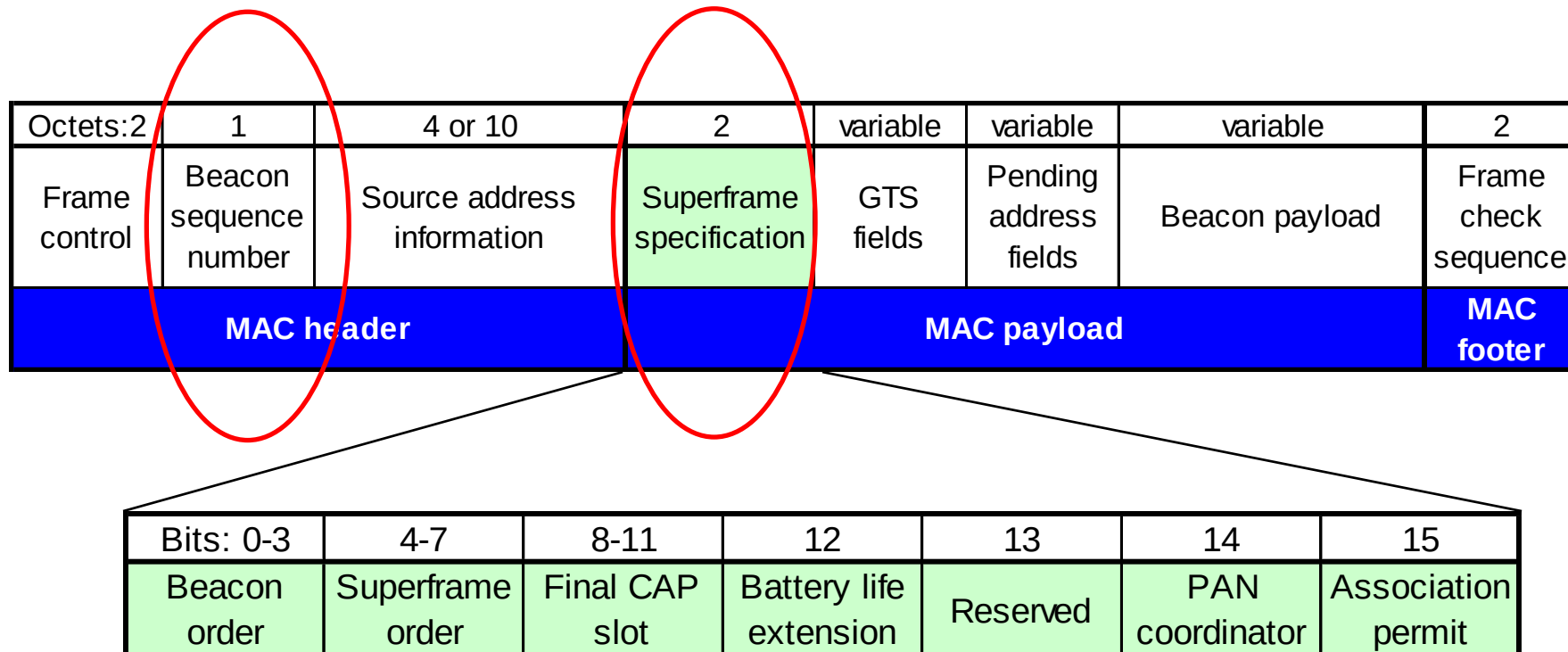
Cont...



Frame Control

Bits: 3	1	1	1	1	3	2	2	2
Frame Type	Security enabled	Frame pending	ACK required	Pan ID	Reserved	Dest addr mode	Frame Version	Src addr mode

Beacon Frame Format



Command Frame Format

Octets:2	1	4 to 20	1	variable	2
Frame control	Data sequence number	Address information	Command type	Command payload	Frame check sequence
MAC header			MAC payload		MAC footer

- **Command Frame Types**

- Association request
- Association response
- Disassociation notification
- Data request
- PAN ID conflict notification
- Orphan Notification
- Beacon request
- Coordinator realignment
- GTS request

Data & ACK Frame Format

Data Frame

Octets:2	1	4 to 20	variable	2
Frame control	Data sequence number	Address information	Data payload	Frame check sequence
MAC header			MAC Payload	MAC footer

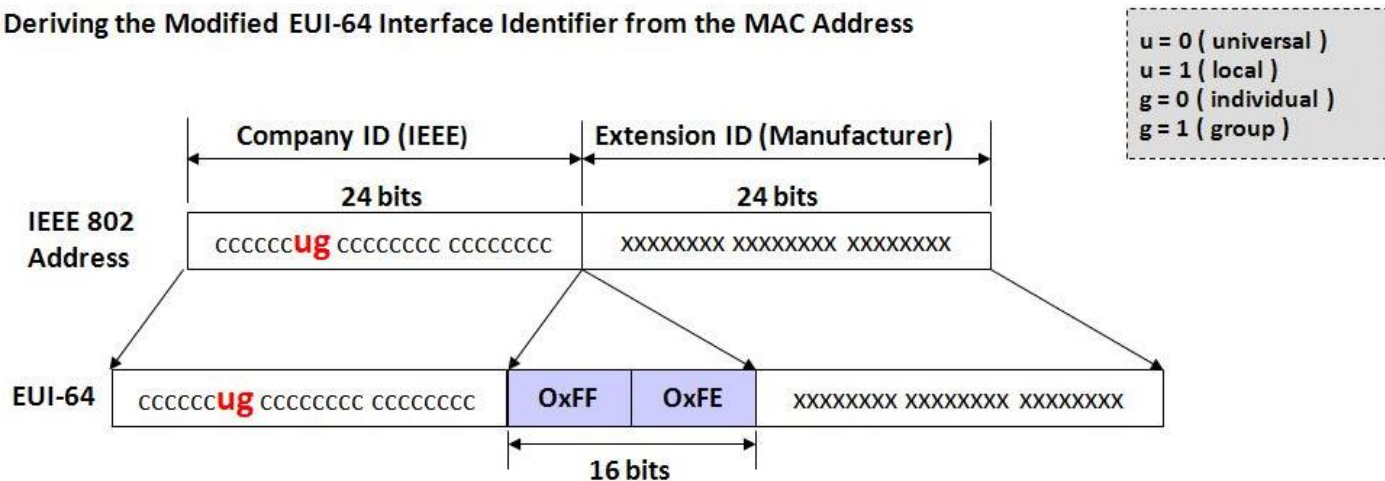
ACK Frame

Octets:2	1	2
Frame control	Data sequence number	Frame check sequence
MAC header		MAC footer

Device Addressing

- Two or more devices communicating on the **same physical channel** constitute a WPAN.
 - A WPAN includes at least one FFD (PAN coordinator)
 - Each independent PAN will select a unique PAN identifier
- Each device operating on a network has a **unique 64-bit address**, called **extended unique identifier (EUI-64)**
 - This address can be used for direct communication in the PAN
- A device also has a **16-bit short address**, which is **allocated by the PAN coordinator** when the device associates with its coordinator.
- IEEE 802.15.4 devices can be grouped into **Personal Area Networks (PANs)**. These are identified by their **2 Byte PAN identifier (PAN ID)**.

Deriving the Modified EUI-64 Interface Identifier from the MAC Address

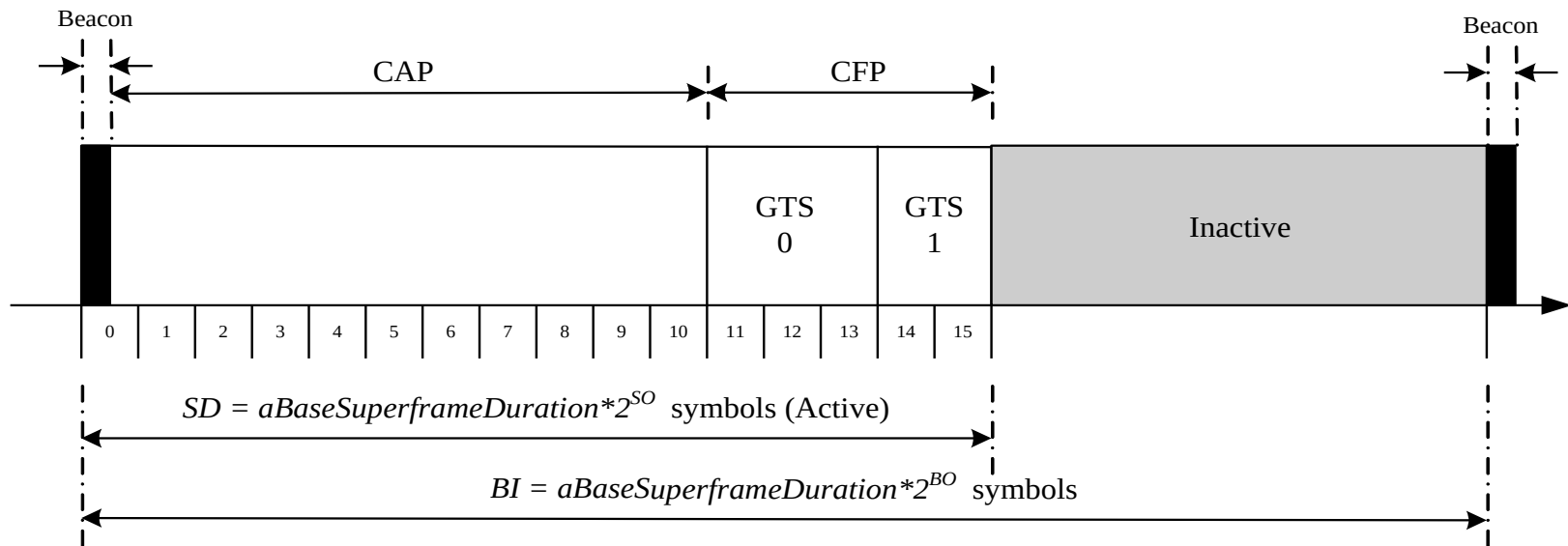


Addressing Modes



- IEEE 802.15.4 frames contain address information for both the source and destination.
- Each of these can be (independently) configured as one of **three different addressing modes**, which sets the address data (none/short/long, with/without PAN ID)
 - **Short address:** The address field includes a short address and a PAN ID (total of 4 bytes).
 - **Long address:** The address field includes a long address and a PAN ID (total of 10 bytes).
 - **No address:**
 - Both addresses are missing from *ACK* frames.
 - For *data* and *command* frames, only one (either source or destination) field can be omitted
 - if the **source address is omitted**, it means the PAN coordinator sent the frame;
 - if the **destination address is missing**, it means it should be received by the PAN coordinator.

Superframe



- A superframe is divided into two parts
 - **Inactive:** all station sleep.
 - no communication
 - nodes can turn their **radios off** and go into power saving mode
 - **Active:**
 - Active period is divided into **16 slots**
 - 16 slots are further divided into two parts
 - Contention access period (**CAP**)
 - Contention free period (**CFP**)
 - Beacon only period (**BOP**)
- **superframe order (SO)** : decides the length of the active portion in a superframe
- **beacon order (BO)** : decides the length of a superframe or beacon transmission period
- **beacon-enabled** network should satisfy $0 \leq SO \leq BO \leq 14$
- PAN coordinator decides SO, BO

Cont...



- *aBaseSlotDuration* = The number of symbols forming a superframe slot when *the superframe order (SO)* is equal to zero = **60 PHY symbols**
- *aBaseSuperframeDuration* = The number of symbols forming a superframe when *the superframe order (SO)* is equal to zero. = ***aBaseSlotDuration* × aNumSuperframeSlots**
- *aNumSuperframeSlots* = The number of slots contained in any superframe = **16**
- Length of a superframe can range from **15.36 msec** to **215.7 sec** (= 3.5 min).
- Beacons are used for
 - starting superframes
 - synchronizing with other devices
 - announcing the existence of a PAN
 - informing pending data in coordinators

Cont...

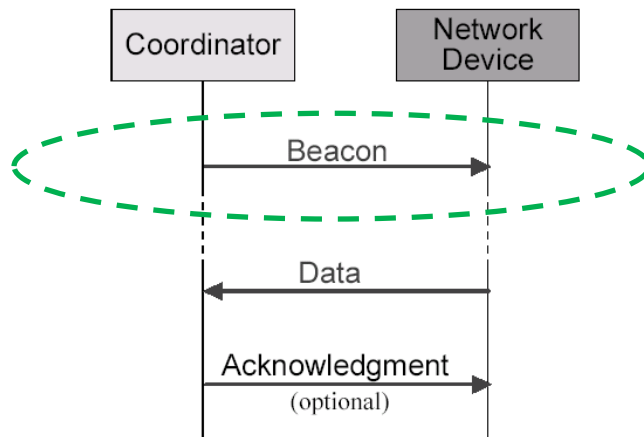


- In a “beacon-enabled” network (i.e. uses superframe structure)
 - Devices use the slotted CAMA/CA mechanism to contend for the channels
 - FFDs who require fixed rates of transmissions can ask for GTS from the coordinator
- In a “nonbeacon-enabled” network (i.e. do not use superframe structure)
 - Devices use the unslotted CAMA/CA mechanism for channel access
 - GTS shall not be permitted
- CSMA/CA is not used for Beacon transmission; also not for Data frame transmission during CFP
- Each device will be
 - active for $2^{-(BO-SO)}$ portion of the time
 - sleep for $1 - 2^{-(BO-SO)}$ portion of the time
- Duty Cycle:

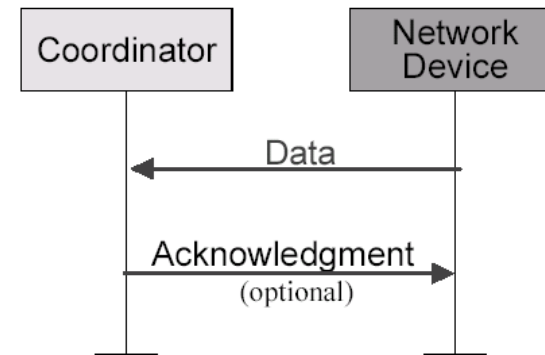
BO-SO	0	1	2	3	4	5	6	7	8	9	≥ 10
Duty cycle (%)	100	50	25	12	6.25	3.125	1.56	0.78	0.39	0.195	< 0.1

Data Transfer: Device -> Coordinator

- Data transferred **from device to coordinator**
 - In a **beacon-enabled network**,
 - a device finds the beacon **to synchronize** to the superframe structure.
 - Then it uses **slotted CSMA/CA** to transmit its data.
 - In a **non-beacon-enabled network**,
 - device simply transmits its data using **unslotted CSMA/CA**



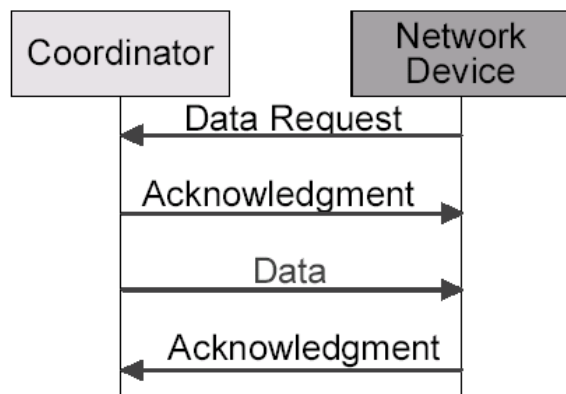
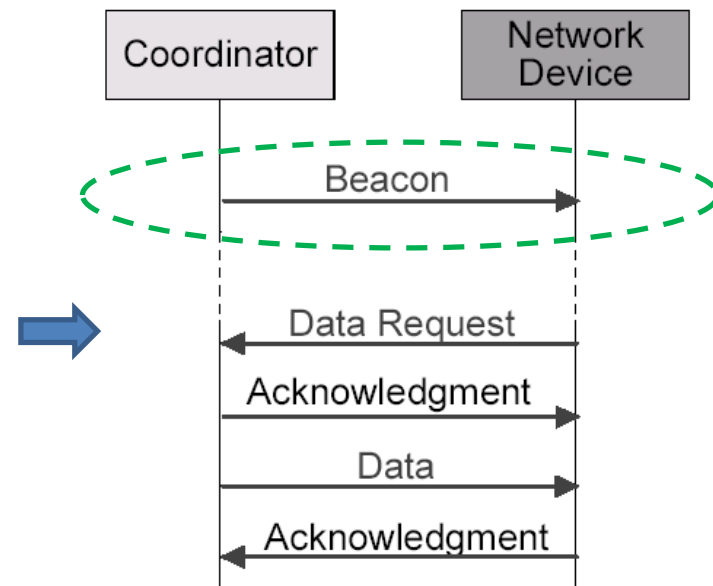
Communication to a coordinator
In a **beacon-enabled** network



Communication to a coordinator
In a **non-beacon-enabled** network

Data Transfer: Coordinator -> Device

- Data transferred **from coordinator to device**
 - in a **beacon-enabled** network:
 - The **coordinator indicates** in the beacon that some data is pending.
 - A device periodically listens to the beacon and transmits a Data Request command using **slotted CSMA/CA**.
 - Then ACK, Data, and ACK



- Data transferred **from coordinator to device**
 - in a **non-beacon-enabled** network:
 - The device transmits a Data Request using **unslotted CSMA/CA**.
 - If the coordinator has its pending data, an ACK is replied.
 - Then the coordinator transmits Data using **unslotted CSMA/CA**.
 - If there is no pending data, a data frame with zero length payload is transmitted.

Channel Access Mechanism

- CSMA/CA random channel access is used more often

➤ beacon-enabled network → uses **slotted CSMA/CA**

In slotted CSMA/CA:

- The **backoff period boundaries** of every **device** in the PAN shall be **aligned with** the **superframe slot boundaries** of the PAN coordinator
 - i.e. the start of first **backoff period** of each device is aligned with the start of the beacon transmission
- The MAC sublayer shall ensure that the PHY layer commences all of its **transmissions on the boundary of a backoff period**

➤ nonbeacon-enabled network → uses **unslotted CSMA/CA**

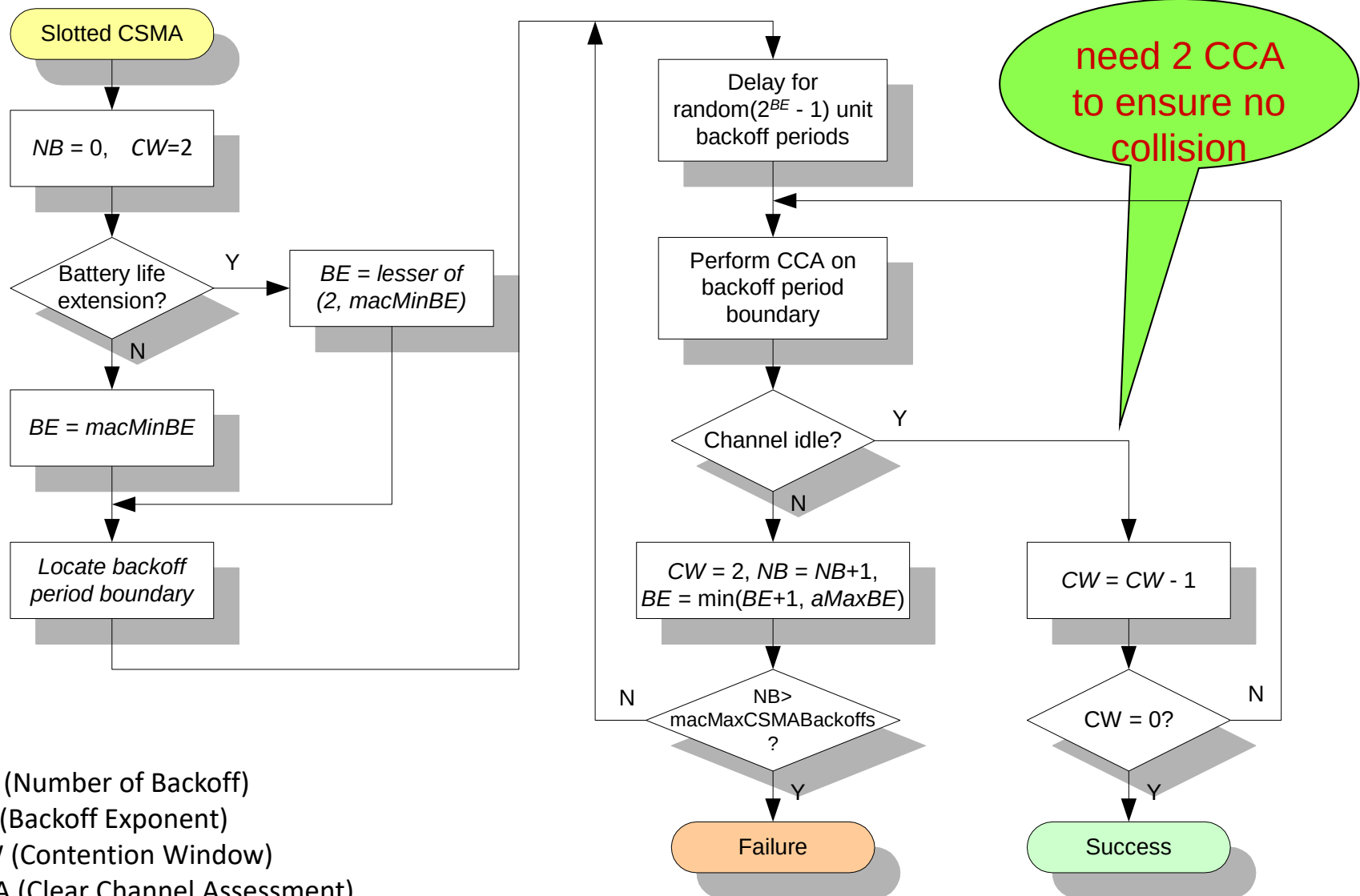
In unslotted CSMA/CA:

- The **backoff periods** of one device **are not related** in time to the backoff periods of any other device in the PAN.
- Algorithms runs using **units of time** called backoff periods, where one backoff period shall be equal to *aUnitBackoffPeriod*.

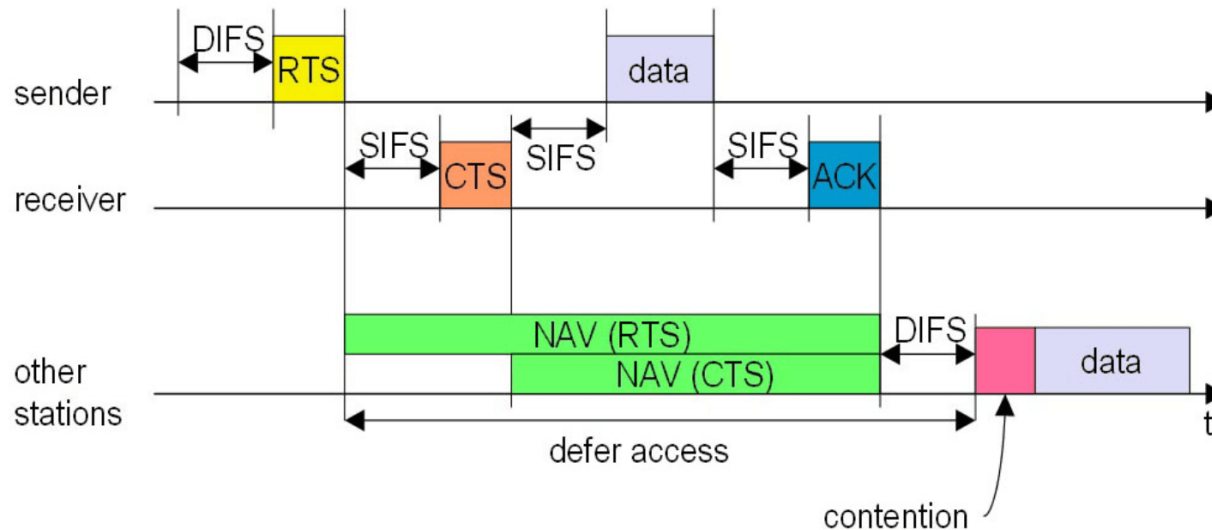
Slotted CSMA/CA



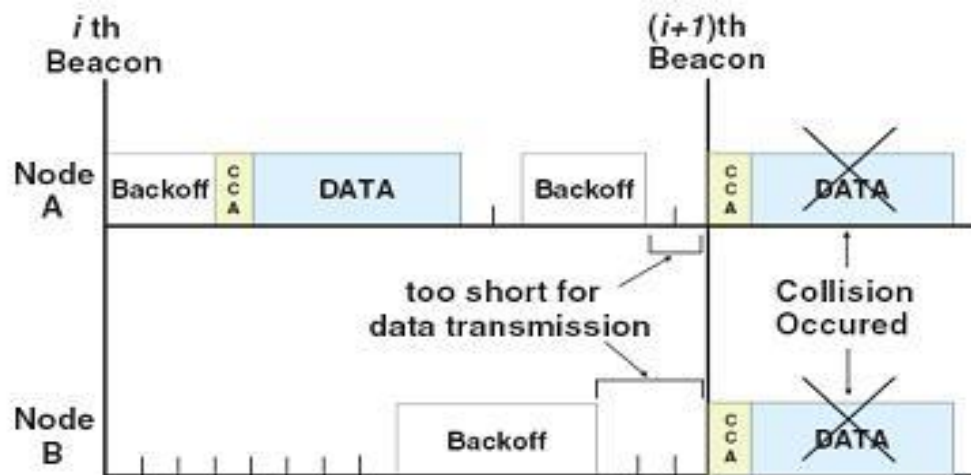
- Each device maintains 3 variables for each transmission attempt
 - **NB (Number of Backoff)**: number of times that backoff has been taken in this attempt of transmission
 - if exceeding **macMaxCSMABackoff**, the attempt fails
 - **BE (Backoff Exponent)**: the backoff exponent is related to **how many backoff periods** a device shall wait before attempting to assess a channel.
 - the number of backoff periods is **greater than** the remaining number of backoff periods in the CAP
 - MAC sublayer shall pause the backoff countdown at the end of the CAP, and resume it at the start of the CAP in the next superframe
 - **CW (Contention Window)**: contention window length, the number clear slots that must be seen after each backoff
 - **always set to 2** and **count down to 0** if the channel is sensed to be clear
 - The design is for some PHY parameters, which require 2 CCA for efficient channel usage.
- **Battery Life Extension (BLE)**:
 - designed for very low-power operation, where a node **only contends in the first few slots**



Contention in 802.11 & 802.15.4



Contention in
IEEE 802.11 DCF



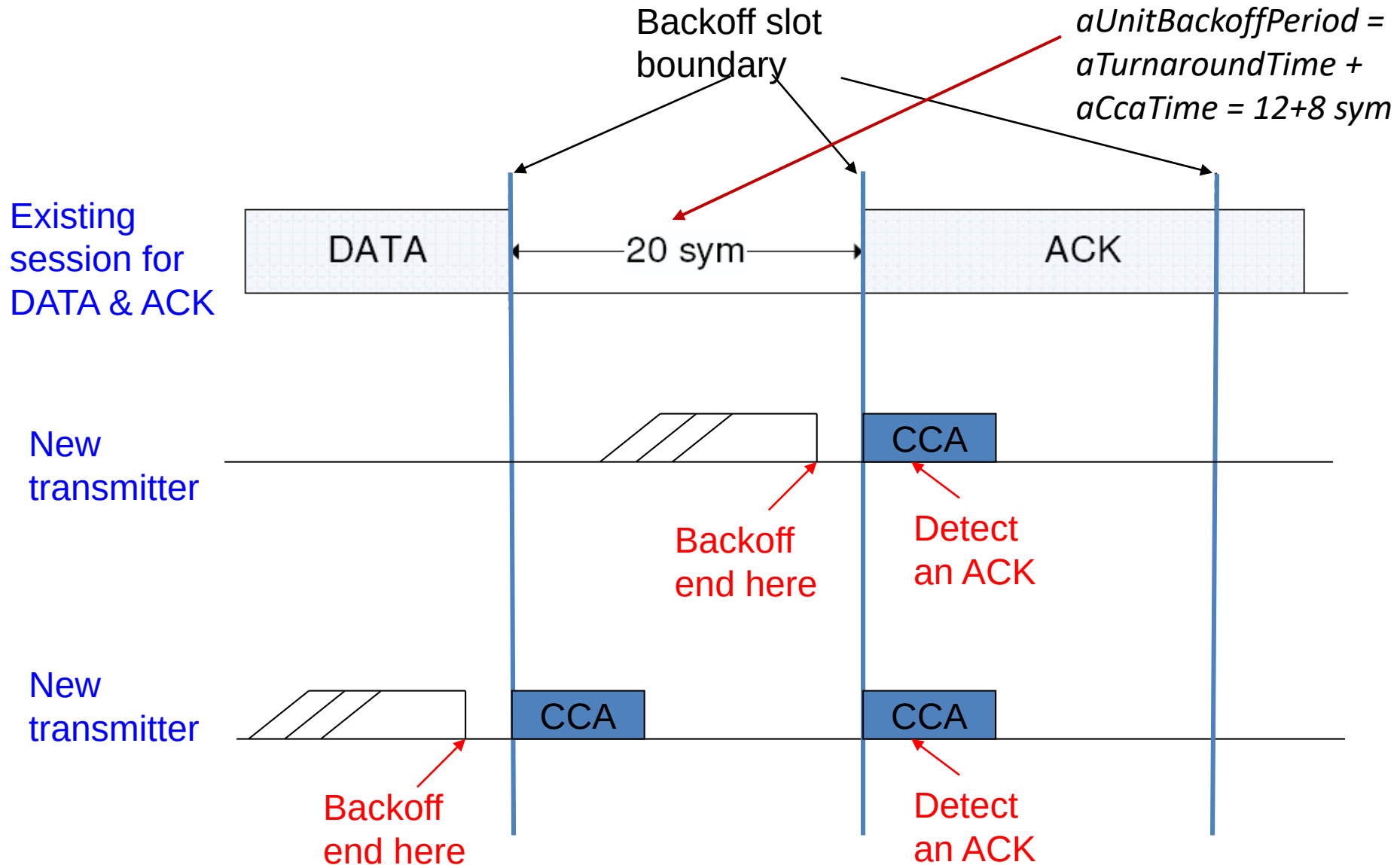
Contention in
IEEE 802.15.4

Why 2 CCAs to Ensure Collision-Free

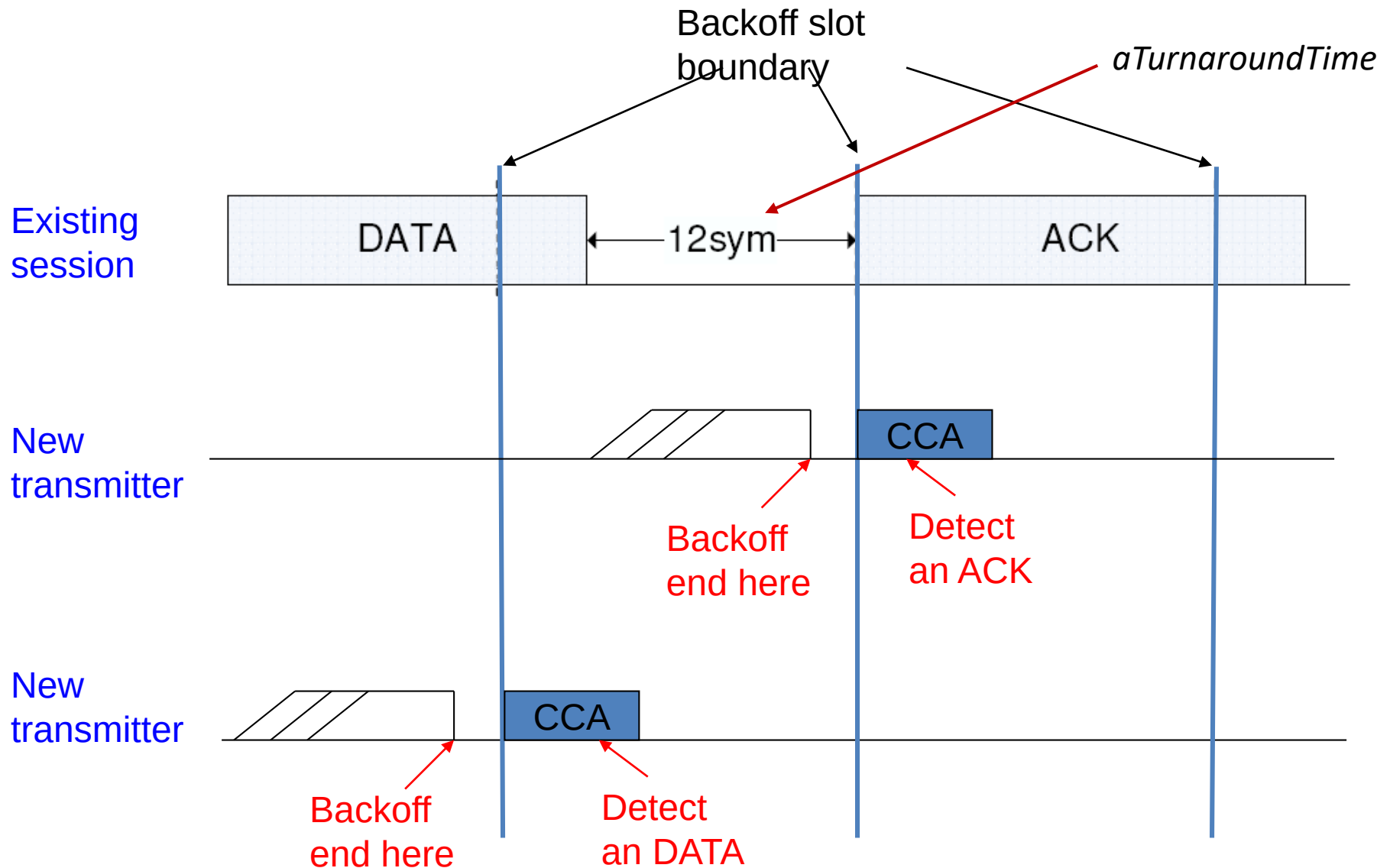


- Each CCA occurs at the boundary of a backoff slot (= 20 PHY symbols), and each CCA time = 8 PHY symbols.
- The standard specifies that a transmitter node performs the CCA twice in order to protect acknowledgment (ACK).
 - When an ACK packet is expected, the receiver shall send it after a t_{ACK} time on the backoff boundary
 - t_{ACK} varies from 12 to 31 symbols
 - One-time CCA of a transmitter may potentially cause a collision between a newly-transmitted packet and an ACK packet.
 - (See examples below)

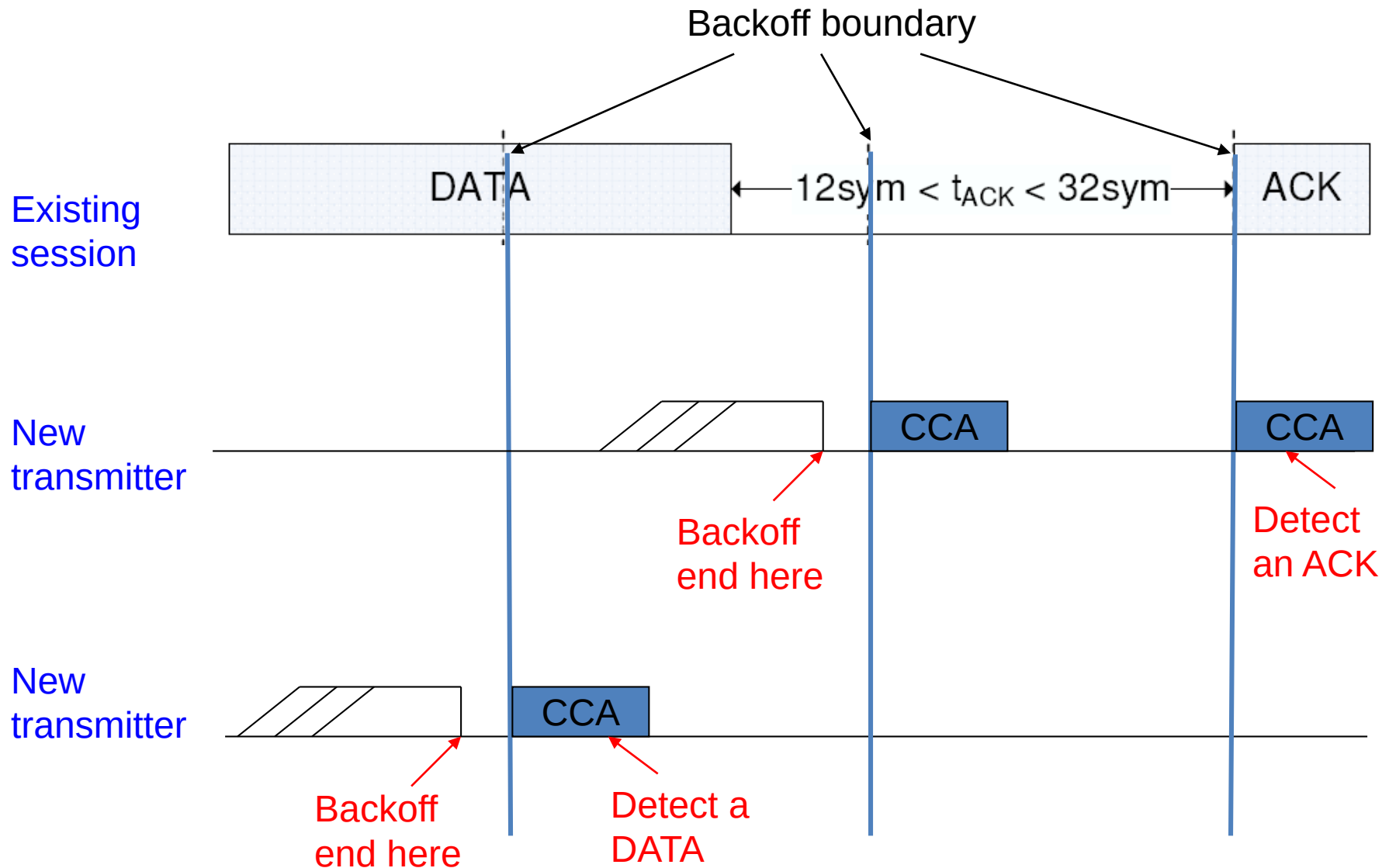
Why 2 CCAs (case 1)



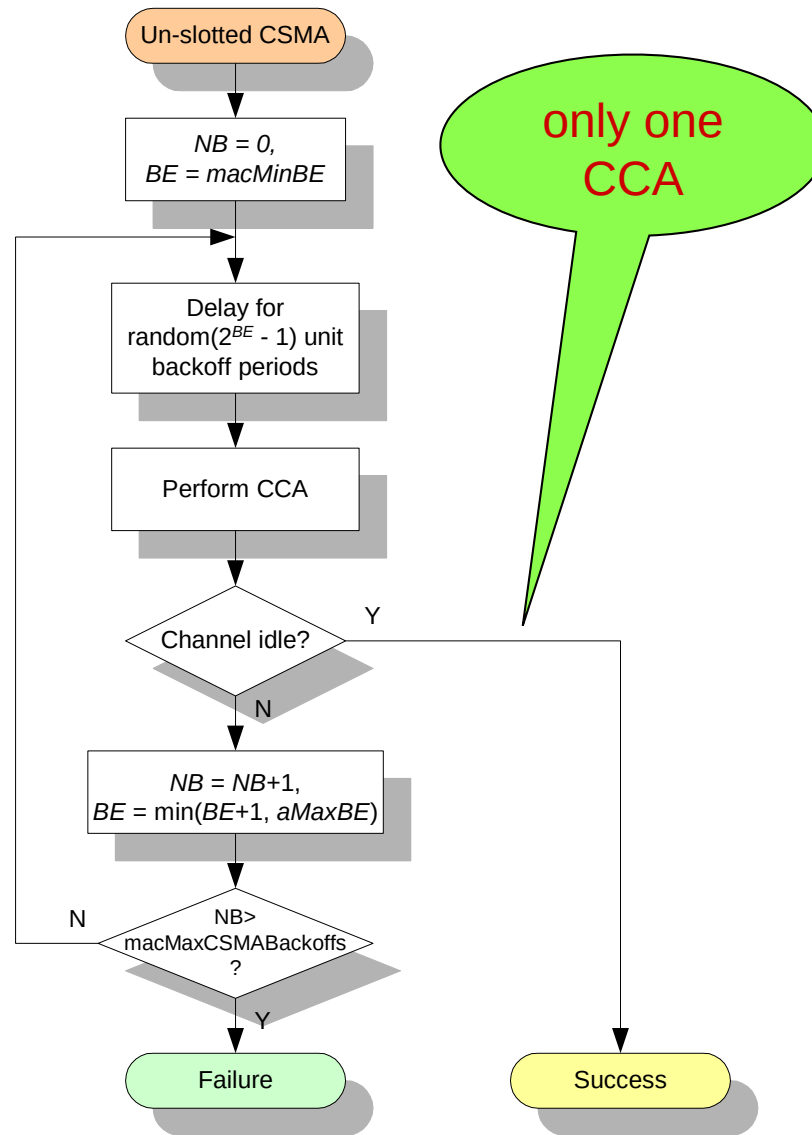
Why 2 CCAs (Case 2)



Why 2 CCAs (Case 3)



Unslotted CSMA/CA



NB (Number of Backoff)
BE (Backoff Exponent)
CW (Contention Window)
CCA (Clear Channel Assessment)

GTS Concepts



- A **guaranteed time slot (GTS)** allows a device to operate on the channel within a portion of the superframe
- A GTS shall only be **allocated by the PAN coordinator**
- The PAN coordinator can allocated up to **7 GTSs** at the same time
- The PAN coordinator decides whether to allocate GTS based on:
 - Requirements of the GTS request
 - The current available capacity in the superframe
- A GTS can be deallocated
 - At any time at the discretion of the PAN coordinator or
 - By the device that originally requested the GTS
- A device that has been allocated a GTS may also operate in the CAP
- A data frame transmitted in an allocated GTS **shall use only short addressing**

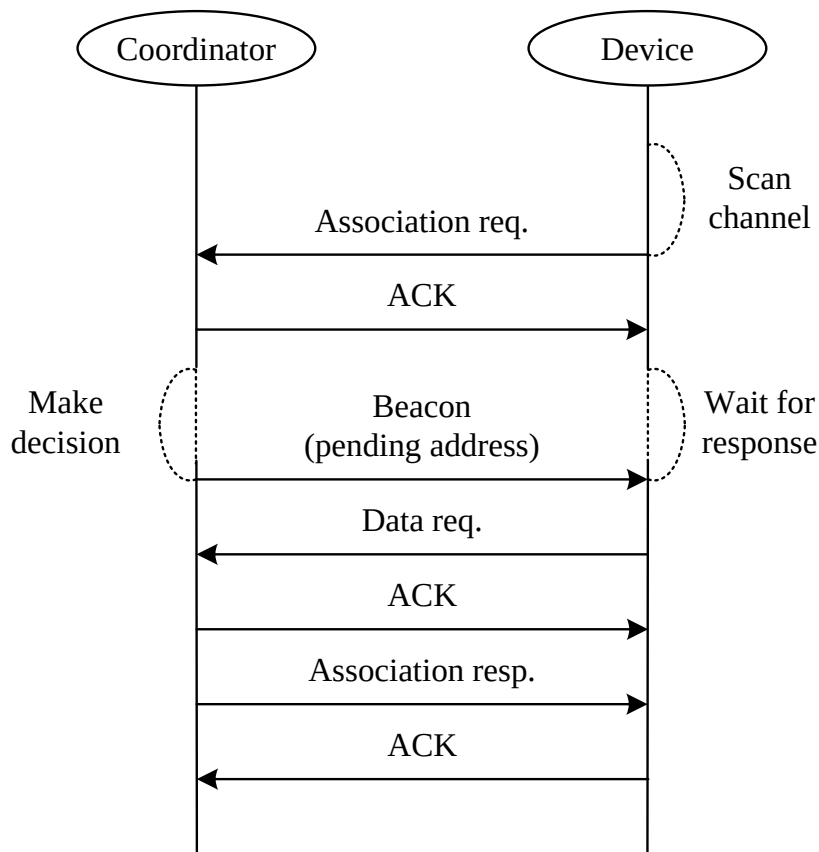
Cont...

- Before GTS starts, the **GTS direction** shall be specified as either transmit or receive
 - Each device may request **one transmit GTS** and/or **one receive GTS**
- A device shall only attempt to allocate and use a GTS if it is currently tracking the beacon
- If a device **loses synchronization** with the PAN coordinator, all its **GTS allocations shall be lost**
- The use of GTSs by an RFD is optional

Association Procedures

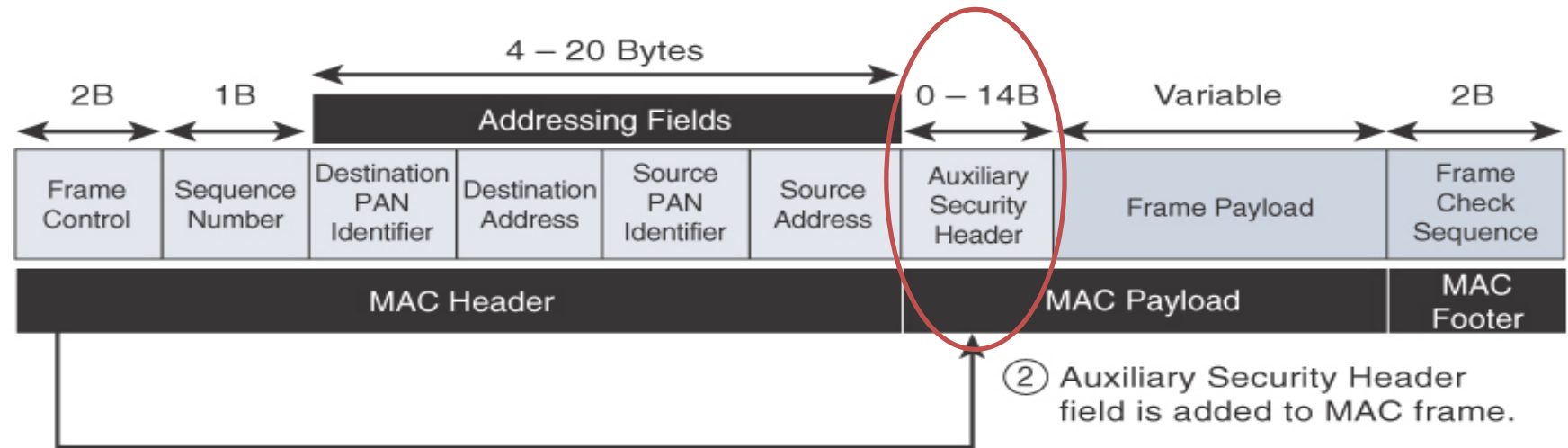
- A device becomes a member of a PAN by associating with its coordinator

- Procedures:**



- The ACK to an Association Request command does not mean that the device has associated.
- In IEEE 802.15.4, association results are announced in an **indirect fashion**.
 - A coordinator responds to Association Requests by appending devices' long addresses in Beacon frames
- Devices need to send a **data request** to the coordinator to **acquire the association result**
- After associating to a coordinator, a device will be assigned a **16-bit short address**.

Security



① Security Enabled bit in Frame Control is set to 1.

- IEEE 802.15.4 specification uses **Advanced Encryption Standard (AES)** with a 128-bit key length as the base encryption algorithm
- **Message integrity code (MIC)**, which is calculated for the entire frame using the same AES key, to validate the data that is sent

Limitations in 802.15.4



- **Disadvantages of IEEE 802.15.4**
 - MAC reliability
 - unbounded latency
 - multipath fading
- **IEEE 802.15.4e** amendment of 802.15.4-2011 expands the MAC layer feature set
 - to remedy the disadvantages of 802.15.4.
 - to better suitable in factory and process automation, and smart grid
 - **Main modifications** were:
 - frame format,
 - security,
 - determinism mechanism, and
 - frequency hopping
- **IEEE 802.15.4g** amendment of 802.15.4-2011 expands mainly PHY layer feature set
 - to optimize large outdoor wireless mesh networks for field area networks (FANs)
 - to better suitable in smart grid or smart utility network (SUN) communication
 - **Main modifications** were:
 - New PHY definitions
 - some MAC modifications needed to support the new PHY

Lessons Learned



- ✓ What is IEEE 802.15.4
- ✓ What is ZigBee
- ✓ IEEE 802.15.4. PHY
 - Functionalities
 - Spread Spectrum, DSSS
 - Modulation, QPSK, OQPSK
- ✓ IEEE 802.15.4 MAC
 - MAC Frame Formats
 - Timeslot, Superframe
 - Device Addressing
 - Data Transfer Model
 - Channel Access Methods
 - Guaranteed time slot (GTS)
 - Association Procedure
 - Security
- ✓ Limitations of IEEE 802.15.4

Thanks!



Figures and slide materials are taken from the following sources:

1. David Hanes *et al.*, “IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”, 1st Edition, 2018, Pearson India.
2. Oliver Hersent et al., “The Internet of Things: Key Applications and Protocols”, 2018, Wiley India Pvt. Ltd.